

Nicolò Bonacorsi

(646) 394-1928 | nb3328@columbia.edu | [linkedin.com/in/nicolobonacorsi](https://www.linkedin.com/in/nicolobonacorsi)

Education

Columbia University

MS in Applied Mathematics

- Teaching Assistant – APMA E4300 Numerical Methods

Sapienza University of Rome

BS in Mathematical Sciences for Artificial Intelligence, GPA: 3.99/4.00

Final Grade: 110/110 cum laude

University of Bologna

BS in Statistical Sciences, GPA: 3.95/4.00

Final Grade: 110/110 cum laude; Award: University of Bologna Prize for Outstanding Student

New York, NY

Aug 2025 – Dec 2026

Jan 2026 – Present

Rome, Italy

Sep 2022 – Sep 2025

Bologna, Italy

Sep 2021 – Jul 2024

Publications

- Nicolò Bonacorsi, Michele Caprio, Francesca Meneghello, and Francesco Restuccia (2026). *Credal Rate-Distortion Theory*. Submitted to IEEE Information Theory Workshop (ITW) 2026. Mentis Laboratory preprint. *Credal Rate-Distortion Theory*.
- Nicolò Bonacorsi and Matteo Bordoni (2026). *Bayesian Modeling of Collatz Stopping Times: A Probabilistic Machine Learning Perspective*. arXiv:2603.04479. <https://arxiv.org/abs/2603.04479>

Research Experience

Columbia University Supervisor: Professor Ioannis Karatzas

Graduate Researcher

- Built a stochastic control and optimal stopping framework for controlled diffusions with endogenous drift and volatility, deriving a tractable shadow-price reduction of the fully nonlinear HJB free-boundary problem.
- Characterized when BGK-type models admit explicit threshold policies with constant optimal controls, and proved rigidity results showing that such closed-form solutions require constant Hamiltonian curvature.
- Developed verification tools for nonconstant-curvature regimes, proving optimality of state-dependent feedback policies and first-hitting stopping rules over the full finite-cost admissible class.

Northeastern University Supervisor: Professor Francesco Restuccia

Graduate Researcher

- Formulated a credal free-energy distortion theory for uncertainty-aware probabilistic compression, proving exact reverse-inclusion, hull-max aggregation, graph-coloring compression, and local block-lift laws governed by OR graph powers and Körner graph entropy.
- Validated the framework on SVHN predictive credal sets built from 3×5 ensemble-view distributions, computing exact chromatic numbers for $N = 60$ inputs and compressing 60 credal likelihoods into 14 admissible codewords at $4.286\times$ compression with 0 credal violations.

Projects

Columbia University

Bayesian Modeling of Collatz Stopping Times

- Built a large-scale Bayesian modeling pipeline for Collatz stopping times with $n \leq 10^7$, fitting hierarchical Negative-Binomial GLMs with arithmetic regime effects ($n \bmod 8$) to model overdispersed and heavy-tailed count data.
- Calibrated a Monte Carlo simulation engine via Bayesian updating; regime conditioning reduced Wasserstein distance to the empirical distribution by $3.2\times$, improving tail-risk representation, posterior calibration, and distributional fit.

Sapienza University of Rome

Spectral Methods for High-Dimensional Prediction and Face Recognition

- Applied PCA via SVD to high-dimensional U.S. crime data, reducing multicollinearity and improving out-of-sample R^2 from 0.5544 to 0.8555 through rank-controlled regression and cross-validation.
- Implemented an eigenfaces-based recognition algorithm using low-rank orthogonal projections, achieving 98.3% classification accuracy while improving numerical stability and robustness to noise through spectral regularization.

University of Bologna

SARIMAX Modeling of Vehicle Registration Dynamics

- Constructed and validated a SARIMAX(p, d, q)(P, D, Q) $_s$ forecasting model with macroeconomic exogenous variables, performing stationarity diagnostics, AIC/BIC model selection, rolling-window validation, and residual Ljung-Box testing.
- Achieved 0.3% ex-post out-of-sample forecasting error, matching a feedforward neural-network benchmark while preserving interpretability, lower variance, and analytical uncertainty quantification.

Professional Experience

EndParty Group

Quantitative Research Lead & Co-Founder

- Built forecasting models for event revenues and sponsorship returns using historical demand, behavioral indicators, and macroeconomic proxies; used model outputs to support capital allocation and event-planning decisions across 4 years of operations.
- Developed data-driven sponsorship analyses quantifying expected ROI for commercial partners; supported pitches that secured 12 agreements and contributed to a $12\times$ revenue increase from the 2019 baseline to 2023.

Skills

Programming / Data: Python, SQL, R, MATLAB, NumPy, pandas, SciPy, scikit-learn, PyTorch, statsmodels.

Quantitative Research: Stochastic Processes, Stochastic Control, Optimal Stopping, Time Series Analysis, Monte Carlo Simulation, Bayesian Inference, Statistical Learning, Risk Modeling.

Machine Learning / Deep Learning: GLMs, Hierarchical Bayesian Models, Neural Networks, Ensemble Methods, Cross-Validation, Feature Engineering, Model Evaluation, Variational Inference.

Numerical / Mathematical Methods: Optimization, Numerical Methods, Differential Equations, HJB Equations, Free-Boundary Problems, PCA/SVD, Information Theory, Uncertainty Quantification.

Certifications: BOOST'24 and FA'24 Certified Student, Elicsir Foundation and Schmidt Sciences.

Languages: Italian (native), English (fluent).